Git & GitHub Exam Project

Version Control Workflow

(Local Git + GitHub)

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February 2026

What This Project Is About:

I created a practice project on my computer, make changes to files, and upload it to GitHub - all while taking screenshots to prove what I did correctly.

Task 3.1: Create Local Repo

I created a folder on my computer and tell Git to start tracking changes in it.

```
C:\Users\peria\OneDrive\Desktop>cd git_exam_Periklis_Andrianos

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git init
Initialized empty Git repository in C:/Users/peria/OneDrive/Desktop/git_exam_Periklis_Andrianos/.git/

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git config user.name peri1387

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git config user.email periandrianos@gmail.com

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

Task 3.2: First Commit

I created my first file (README.md) and saved a "snapshot" of it in Git. This snapshot is called a "commit."

```
C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git add README.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git commit -m "Initial commit"
[master (root-commit) b2d1afd] Initial commit
 1 file changed, 7 insertions(+)
 create mode 100644 README.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git log --oneline
b2d1afd (HEAD -> master) Initial commit
```

What i did :

* I included this file in my next snapshot.

* I saved the snapshot with a commit message.

* View my commit history.

Task 3.3: .gitignore

Told the Git to ignore a temporary file using a special file called .gitignore . That's the way to have untracked files on your Git.

```
C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        temp.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>echo temp.txt > .gitignore

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)
```

Git now ignores temp.txt completely. It won't track it, won't show it in status. The .gitignore file is like a list that says "don't pay attention to these files."

Task 3.4: Branch Workflow

Branches let us work on new features without messing up our main work.

It is like creating a copy where you can experiment safely.

```
C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git branch feature/notes

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git checkout feature/notes
Switched to branch 'feature/notes'

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git add notes.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git commit -m "Add project notes"
[feature/notes 94daf73] Add project notes
 1 file changed, 14 insertions(+)
 create mode 100644 notes.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git log --oneline --decorate
94daf73 (HEAD -> feature/notes) Add project notes
b2d1afd (master) Initial commit
```

What I did:

* I created a separate branch called feature/notes.

* I switched to it and made changes there.

* My master branch is untouched - the notes.md file doesn't exist there yet!

Task 3.5: Merge Conflict

A merge conflict happens when the same line in a file is changed differently in two branches.

So I edited the same line in README.md on master and on feature/notes.

When I tried to merge feature/notes into master , I created the conflict.

```
C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git checkout master
Switched to branch 'master'

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git add README.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git commit -m "Update title on main branch"
[master 99edc34] Update title on main branch
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git checkout feature/notes
Switched to branch 'feature/notes'

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git add README.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git commit -m "Update title on feature branch"
[feature/notes 75aeff5] Update title on feature branch
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git checkout main
error: pathspec 'main' did not match any file(s) known to git

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git merge feature/notes
Already up to date.

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git checkout master
Switched to branch 'master'

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git merge feature/notes
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
```

So I tried to resolve this . I decided which version to keep. I removed the markers and kept the correct final text.

```
README.md !
C: > Users > peria > OneDrive > Desktop > git_exam_Periklis_Andrianos > README.md > # <<<<<<< HEADMy Git Exam Project - Main Branch Version
You, 5 minutes ago | 1 author (You) | Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
1  <<<<<<< HEAD (Current Change)
2  **My Git Exam Project - Main Branch Version**
3  =======
4  **My Git Exam Project Feature Branch Version**
5  >>>>>>> feature/notes (Incoming Change)
6
7   
8
9  * This is my first Git project
10 * I am learning version control
11 * This project demonstrates Git workflow
12
```

Then I went back to the terminal to commit the resolution.

```
C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git add README.md

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git status
On branch master
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
        modified:   README.md
        new file:   notes.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore


C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git commit -m "Resolve merge conflict"
[master c2ad55f] Resolve merge conflict

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git log --oneline
c2ad55f (HEAD -> master) Resolve merge conflict
75aeff5 (feature/notes) Update title on feature branch
99edc34 Update title on main branch
94daf73 Add project notes
b2d1afd Initial commit
```

Task 3.6: Push to GitHub

It's time to upload my local Git project to GitHub so it's stored online and others can see it.

What I did first:

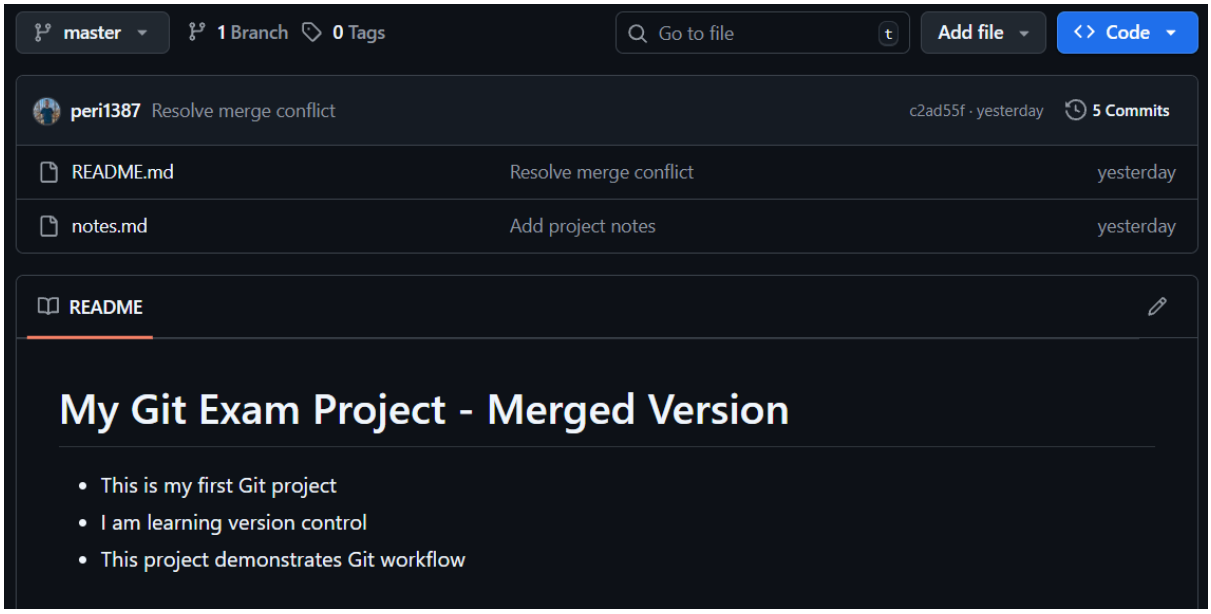
* Went to GitHub.

* Created a new repository.

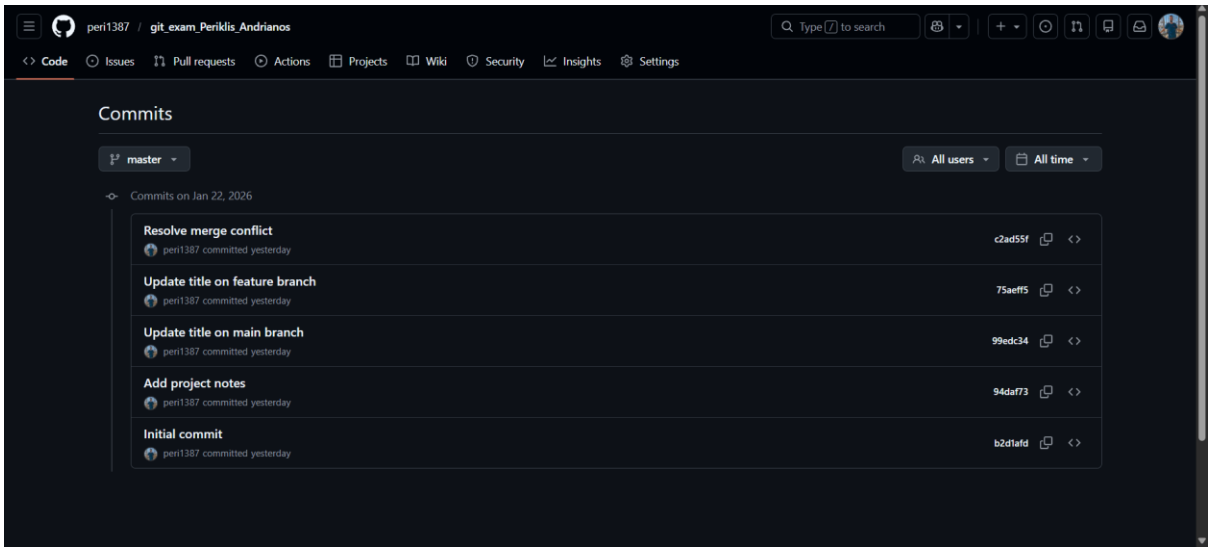
* Connected my Local Repo to GitHub with URL from GitHub.

```
C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git remote add origin https://github.com/peri1387/git_exam_Periklis_Andrianos.git

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git remote -v
origin  https://github.com/peri1387/git_exam_Periklis_Andrianos.git (fetch)
origin  https://github.com/peri1387/git_exam_Periklis_Andrianos.git (push)

C:\Users\peria\OneDrive\Desktop\git_exam_Periklis_Andrianos>git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 12 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (15/15), 1.52 KiB | 520.00 KiB/s, done.
Total 15 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/peri1387/git_exam_Periklis_Andrianos.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Verified my new repository on GitHub.



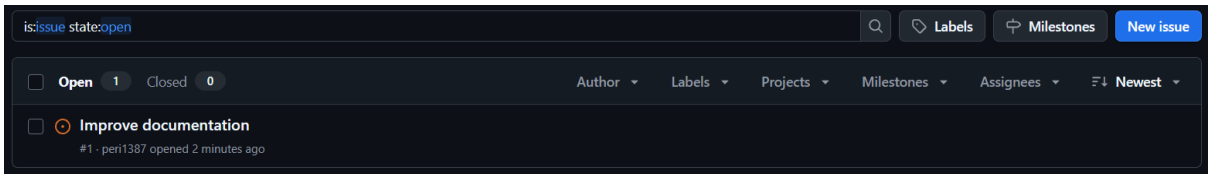
And this is my GitHub page showing my commits history.



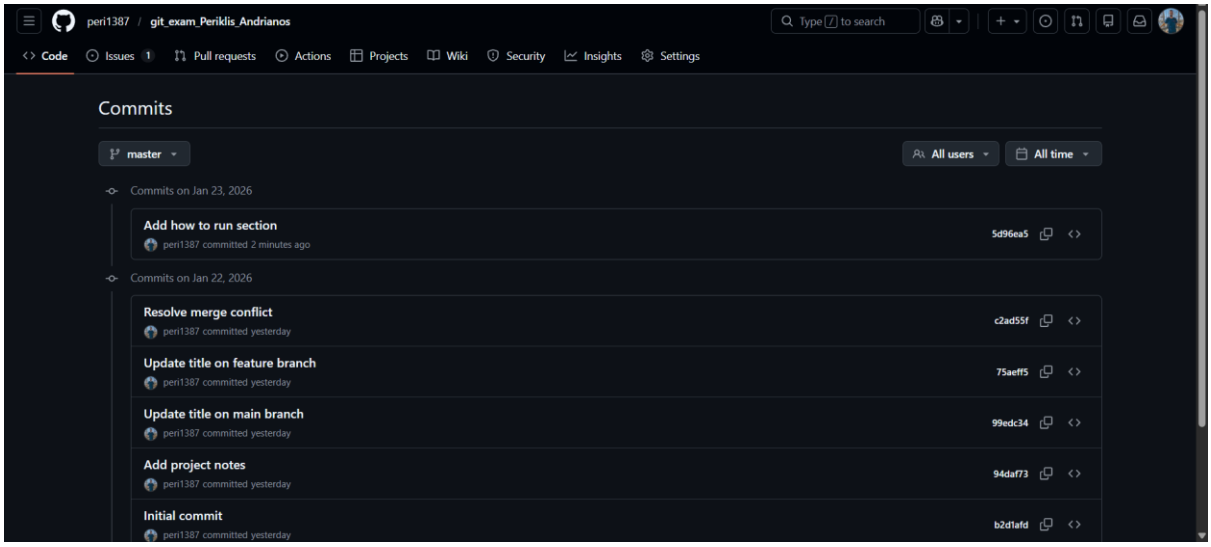
Task 3.7: GitHub Issue + Final Push

GitHub Issues are like a to-do list for our project.

I created an issue, then made an improvement to my README and pushed it.



Here is my GitHub commits updated :



This is my link for the issue :

https://github.com/peri1387/git_exam_Periklis_Andrianos/issues/1